

ChE 312 – Transport Processes II

II. Fundamentals of mass transfer.

II.c. Diffusivities ("diffusion coefficients") from data, correlation, and theory.

- Examine diffusivity variations, magnitudes, and trends.

II.d. Mass-transfer equivalents of the Navier-Stokes equations.

- The continuity equation(s) for mass transfer.
- Need initial conditions and boundary conditions for mass transfer (diffusion or convective mass transfer to an interface).
- De-dimensionalize species-conservation equations and boundary conditions.
 - De-dimensionalizing gives useful dimensionless parameters.

Fundamental relations to remember.

- N_A (total flux) = J_A (diffusion flux) + vC_A (convective flux)

- Fick's Law of Diffusion (1855):

$$J_A = -D_{AB} \frac{dC_A}{dz}$$

- Same form as Fourier's Law of Conduction:

$$\frac{q}{A} = -k \frac{dT}{dz}$$

- Bulk convection of species A:

$$vC_A = \left(\frac{\sum N_i}{C} \right) C_A$$

- Components of total flux N_A :

$$N_A = vC_A + J_A \xrightarrow{\text{binary}} \left(\frac{N_A + N_B}{C} \right) C_A - D_{AB} \frac{dC_A}{dz}$$

- Ref. Case 1: Equimolar counterdiffusion

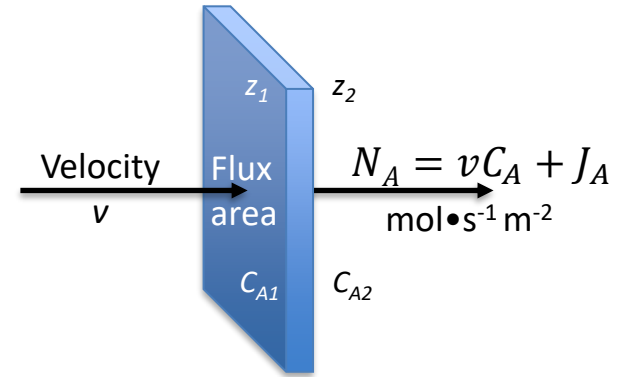
$$N_A = \left(\frac{D_{AB}}{\Delta z} \right) \Delta C_A$$

- Ref. Case 2: A through nondiffusing B

$$N_A = \left(\frac{1}{(x_B)_{\log \text{ mean}}} \frac{D_{AB}}{\Delta z} \right) \Delta C_A \text{ where } (x_B)_{lm} = \frac{x_{B2} - x_{B1}}{\ln(x_{B2}/x_{B1})}$$

- Convective MT to/from a surface:

$$N_{A,\text{convective}} = k \cdot (C_{A,\text{bulk}} - C_{A,\text{surface}})$$



Review data tables,
examining trends
and magnitudes:
First, gas-phase
diffusivities...

System	Temperature		Diffusivity $[(\text{m}^2/\text{s})10^4 \text{ or } \text{cm}^2/\text{s}]$	Ref.
	$^{\circ}\text{C}$	K		
Air-NH ₃	0	273	0.198	(W1)
Air-H ₂ O	0	273	0.220	(N2)
	25	298	0.260	(L1)
	42	315	0.288	(M1)
Air-CO ₂	3	276	0.142	(H1)
	44	317	0.177	
Air-H ₂	0	273	0.611	(N2)
Air-C ₂ H ₅ OH	25	298	0.135	(M1)
	42	315	0.145	
Air-CH ₃ COOH	0	273	0.106	(N2)
Air- <i>n</i> -hexane	21	294	0.080	(C1)
Air-benzene	25	298	0.0962	(L1)
Air-toluene	25.9	298.9	0.086	(G1)
Air- <i>n</i> -butanol	0	273	0.0703	(N2)
	25.9	298.9	0.087	
H ₂ -CH ₄	25	298	0.726	(C2)
H ₂ -N ₂	25	298	0.784	(B1)
	85	358	1.052	
H ₂ -benzene	38.1	311.1	0.404	(H2)
H ₂ -Ar	22.4	295.4	0.83	(W2)
H ₂ -NH ₃	25	298	0.783	(B1)
H ₂ -SO ₂	50	323	0.61	(S1)
H ₂ -C ₂ H ₅ OH	67	340	0.586	(T1)
He-Ar	25	298	0.729	(S2)
He- <i>n</i> -butanol	150	423	0.587	(S2)
He-air	44	317	0.765	(H1)
He-CH ₄	25	298	0.675	(C2)
He-N ₂	25	298	0.687	(S2)
He-O ₂	25	298	0.729	(S2)

Compare dilute-liquid diffusivities:
 $\sim 10^5$ lower, but
 concentration C is
 $\sim 10^3$ higher.

(Gas is $1/22,414$
 mol/cm^3 at STP, vs.
 1 g/cm^3 for H_2O or
 $1/18 \text{ mol/cm}^3$).

Proportional to
 absolute T !

Solute	Solvent	Temperature		Diffusivity [$(\text{m}^2/\text{s})10^9$ or $(\text{cm}^2/\text{s})10^5$]	Ref.
		$^{\circ}\text{C}$	K		
NH_3	Water	12	285	1.64	(N2)
		15	288	1.77	
O_2	Water	18	291	1.98	(N2)
		25	298	2.41	(V1)
CO_2	Water	25	298	2.00	(V1)
H_2	Water	25	298	4.8	(V1)
Methyl alcohol	Water	15	288	1.26	(J1)
Ethyl alcohol	Water	10	283	0.84	(J1)
		25	298	1.24	(J1)
<i>n</i> -Propyl alcohol	Water	15	288	0.87	(J1)
Formic acid	Water	25	298	1.52	(B4)
Acetic acid	Water	9.7	282.7	0.769	(B4)
		25	298	1.26	(B4)
Propionic acid	Water	25	298	1.01	(B4)
HCl (9 g mol/liter) (2.5 g mol/liter)	Water	10	283	3.3	(N2)
		10	283	2.5	(N2)
Benzoic acid	Water	25	298	1.21	(C4)
Acetone	Water	25	298	1.28	(A2)
Acetic acid	Benzene	25	298	2.09	(C5)
Urea	Ethanol	12	285	0.54	(N2)
Water	Ethanol	25	298	1.13	(H4)
KCl	Water	25	298	1.870	(P2)
KCl	Ethylene glycol	25	298	0.119	(P2)

“Stokes-Einstein Equation”

For infinitely dilute spherical molecules in solvent:

$$D_{AB} = \frac{9.96 \times 10^{-16} T}{\mu V_A^{1/3}}$$

With D_{AB} in m^2/s ,

T in K,

μ in $\text{Pa}\cdot\text{s}$,

V_A , the solute molar volume at its normal boiling point, m^3/kmol

For large biological solute A in aqueous solution (solvent B), “Polson eqn” is easier:

$$D_{AB} = \frac{9.40 \cdot 10^{-15} \cdot T}{\mu M_A^{1/3}}$$

For solutes in liquids:

Wilke-Chang correlation for solute A in solvent B:

$$D_{AB} = \frac{7.4 \times 10^{-8} \sqrt{\phi_B M_B T}}{\mu_B v_A^{0.6}}$$

with D_{AB} cm^2/s , T K, M_B g/mol, μ_B in centipoise;
 v_A , the solute molar volume at its normal B.P., cm^3/mol ;
 ϕ_B “solvent association factor” (H_2O 2.6, CH_3OH 1.9)

TABLE 3.3

Molecular Volumes of Dissolved Light Gases and Atomic Contributions for Other Molecules at the Normal Boiling Point

Source: G. Le Bas, *The Molecular Volumes of Liquid Chemical Compounds*, David McKay, New York (1915).

For solutes in liquids, use Wilke-Chang correlation :

for solute A in solvent B:

$$D_{AB} = \frac{7.4 \times 10^{-8} \sqrt{\phi_B M_B T}}{\mu_B \nu_A^{0.6}}$$

with D_{AB} cm²/s, T K, M_B g/mol, μ_B in centipoise;
 ν_A , the solute molar volume at its normal B.P., cm³/mol;
 ϕ_B “solvent association factor” (H₂O 2.6, methanol 1.9,
 ethanol 1.5, non-associating solvents 1.0).

Table 18.2-4. Atomic and Molar Volumes at the Normal Boiling Point

<i>Material</i>	<i>Atomic Volume (m³/kg mol) 10³</i>	<i>Material</i>	<i>Atomic Volume (m³/kg mol) 10³</i>
C	14.8	Ring, 3-membered	−6
H	3.7	as in ethylene oxide	
O (except as below)	7.4		
Doubly bound as carbonyl	7.4	4-membered 5-membered	−8.5 −11.5
Coupled to two other elements		6-membered Naphthalene ring	−15 −30
In aldehydes, ketones	7.4	Anthracene ring	−47.5
1 methyl esters	9.1		
In methyl ethers	9.9		

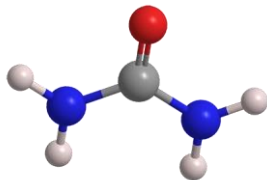
Predict
diffusivities
for
electrolytes
in liquids:

The well-known Nernst–Haskell equation for dilute, single-salt solutions can be used at 25°C to predict the overall diffusivity D_{AB} of the salt A in the solvent B

$$D_{AB}^o = 8.928 \times 10^{-10} T \frac{(1/n_+ + 1/n_-)}{(1/\lambda_+ + 1/\lambda_-)}$$

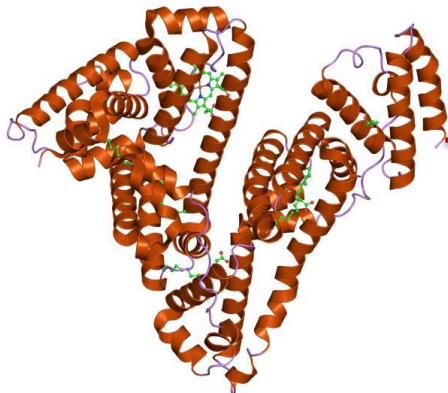
where D_{AB}^o is in cm^2/s , n_+ is the valence of the cation, n_- is the valence of the anion, and λ_+ and λ_- are the limiting ionic conductances in very dilute solutions in $(\text{A}/\text{cm}^2)(\text{V}/\text{cm})(\text{g equiv.}/\text{cm}^2)$.

Limiting Ionic Conductances in Water at 25°C			
Anion	λ_-	Cation	λ_+
OH ⁻	197.6	H ⁺	349.8
Cl ⁻	76.3	Li ⁺	38.7
Br ⁻	78.3	Na ⁺	50.1
NO ₃ ⁻	71.4	K ⁺	73.5
CH ₃ CO ₂ ⁻	40.9	NH ₄ ⁺	73.4
$\frac{1}{2}\text{SO}_4^{2-}$	80	$\frac{1}{2}\text{Ca}^{2+}$	59.5
CLO ₄ ⁻	68.0	$\frac{1}{2}\text{Zn}^{2+}$	53
		$\frac{1}{3}\text{La}^{3+}$	69.5
Ionic conductances in $(\text{A}/\text{cm}^2)(\text{V}/\text{cm})(\text{g equiv.}/\text{cm}^2)$			



Consider diffusivity of biologically important solutes.

The diffusion coefficients for the large protein molecules are on the order of magnitude of $5 \times 10^{-11} \text{ m}^2/\text{s}$ compared to the values of about $1 \times 10^{-9} \text{ m}^2/\text{s}$ for the small solutes ---- This means macromolecules diffuse at a rate about 20 times as slow as small solute molecules for the same concentration differences.



Diffusion Coefficients for Dilute Biological Solutes in Aqueous Solution

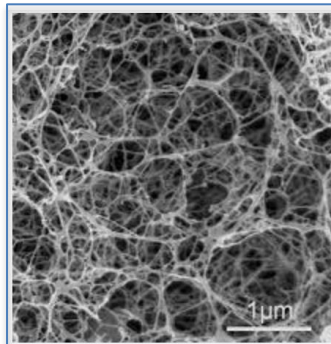
Solute	Temperature		Diffusivity (m^2/s)	Molecular Weight	Ref.
	$^{\circ}\text{C}$	K			
Urea	20	293	1.20×10^{-9}	60.1	(N2)
	25	298	1.378×10^{-9}		(G5)
Glycerol	20	293	0.825×10^{-9}	92.1	(G3)
Glycine	25	298	1.055×10^{-9}	75.1	(L3)
Sodium caprylate	25	298	8.78×10^{-10}	166.2	(G6)
Bovine serum albumin	25	298	6.81×10^{-11}	67 500	(C6)
Urease	25	298	4.01×10^{-11}	482 700	(C7)
	20	293	3.46×10^{-11}		(S6)
Soybean protein	20	293	2.91×10^{-11}	361 800	(S6)
Lipoxidase	20	293	5.59×10^{-11}	97 440	(S6)
Fibrinogen, human	20	293	1.98×10^{-11}	339 700	(S6)
Human serum albumin	20	293	5.93×10^{-11}	72 300	(S6)
γ -Globulin, human	20	293	4.00×10^{-11}	153 100	(S6)
Creatinine	37	310	1.08×10^{-9}	113.1	(C8)
Sucrose	37	310	0.697×10^{-9}	342.3	(C8)
	20	293	0.460×10^{-9}		

Consider diffusivity of biological nutrients in gels: Diffusion through a porous structure.

At right, agar gel used in cell-
growth media, chromatography,
and gel electrophoresis.

If a structure is open enough,
diffusivity is proportional to
% porosity (void fraction ϵ), so

$$D_{effective,AB} = \epsilon \cdot D_{AB}.$$



Solute	Gel	Wt % Gel in Solution	Temperature		Diffusivity (m ² /s)
			K	°C	
Sucrose	Gelatin	0	278	5	0.285 × 10 ⁻⁹
		3.8	278	5	0.209 × 10 ⁻⁹
		10.35	278	5	0.107 × 10 ⁻⁹
		5.1	293	20	0.252 × 10 ⁻⁹
Urea	Gelatin	0	278	5	0.880 × 10 ⁻⁹
		2.9	278	5	0.644 × 10 ⁻⁹
		5.1	278	5	0.609 × 10 ⁻⁹
		10.0	278	5	0.542 × 10 ⁻⁹
		5.1	293	20	0.859 × 10 ⁻⁹
Methanol	Gelatin	3.8	278	5	0.626 × 10 ⁻⁹
Urea	Agar	1.05	278	5	0.727 × 10 ⁻⁹
		3.16	278	5	0.591 × 10 ⁻⁹
		5.15	278	5	0.472 × 10 ⁻⁹
Glycerin	Agar	2.06	278	5	0.297 × 10 ⁻⁹
		6.02	278	5	0.199 × 10 ⁻⁹
Dextrose	Agar	0.79	278	5	0.327 × 10 ⁻⁹
Sucrose	Agar	0.79	278	5	0.247 × 10 ⁻⁹
Ethanol	Agar	5.15	278	5	0.393 × 10 ⁻⁹
NaCl (0.05 M)	Agarose	0	298	25	1.511 × 10 ⁻⁹
		2	298	25	1.398 × 10 ⁻⁹

Diffusivity and permeability through porous media are related.

Diffusion/diffusivity in gas mixtures: The Stefan-Maxwell equation.

- In gases, D_{AB} is normally assumed to be independent of composition.
- Multi-component diffusion can be described best by the Stefan-Maxwell Equation:

$$\frac{dx_i}{dz} = \sum_{j=1}^n \frac{c_i c_j}{c^2 D_{ji}} \left(\frac{J_j}{c_j} - \frac{J_i}{c_i} \right)$$

- A useful simplification of this equation is Blanc's Law for mixture diffusivity in dilute gases.
 - If dilute i diffuses into a homogeneous mixture, then $J_j \cong 0$, and with $\frac{c_j}{c} = x_j$:

$$\frac{dx_i}{dz} = \sum_{j=1}^n \frac{c_i c_j}{c^2 D_{ji}} \left(\frac{J_j}{c_j} - \frac{J_i}{c_i} \right) = -J_i \sum_{j=1, j \neq i}^n \frac{x_j}{c D_{ji}}$$

$$D_{i,mix} \equiv \frac{-J_i}{dx_i/dz} = \left(\sum_{j=1, j \neq i}^n \frac{x_j}{D_{ij}} \right)^{-1}$$

Compare **dilute-liquid diffusivities**: $\sim 10^5$ lower vs gases, but concentration C is $\sim 10^3$ higher.

(Gas is $1/22414 \text{ mol/cm}^3$ at STP, vs. $1 \text{ g/cm}^3 \text{ H}_2\text{O}$, $1/18 \text{ mol/cm}^3$).

~Proportional to absolute T.

<i>Solute</i>	<i>Solvent</i>	<i>Temperature</i>		<i>Diffusivity</i>
		$^{\circ}\text{C}$	K	$[(\text{m}^2/\text{s})10^9 \text{ or } (\text{cm}^2/\text{s})10^5]$
NH_3	Water	12	285	1.64
		15	288	1.77
O_2	Water	18	291	1.98
		25	298	2.41
CO_2	Water	25	298	2.00
H_2	Water	25	298	4.8
Methyl alcohol	Water	15	288	1.26
Ethyl alcohol	Water	10	283	0.84
		25	298	1.24
<i>n</i> -Propyl alcohol	Water	15	288	0.87
Formic acid	Water	25	298	1.52
Acetic acid	Water	9.7	282.7	0.769
		25	298	1.26
Propionic acid	Water	25	298	1.01
HCl (9 g mol/liter)	Water	10	283	3.3
		10	283	2.5
(2.5 g mol/liter)				
Benzoic acid	Water	25	298	1.21
Acetone	Water	25	298	1.28
Acetic acid	Benzene	25	298	2.09
Urea	Ethanol	12	285	0.54
Water	Ethanol	25	298	1.13
KCl	Water	25	298	1.870
KCl	Ethylene glycol	25	298	0.119

For solutes in liquids:

Stokes-Einstein Equation
For infinitely dilute spherical
molecules in solvent:

$$D_{AB} = \frac{9.96 \times 10^{-16} T}{\mu V_A^{1/3}}$$

With D_{AB} in m^2/s ,
 T in K,
 μ in $\text{Pa}\cdot\text{s}$,
 V_A , the solute molar volume at its
normal boiling point, m^3/kmol

For large biological solute A in
aqueous solution (solvent B);
Polson eqn is easier:

$$D_{AB} = \frac{9.40 \cdot 10^{-15} \cdot T}{\mu M_A^{1/3}}$$

Wilke-Chang correlation for solute A in solvent B:

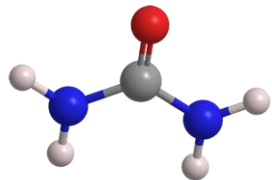
$$D_{AB} = \frac{7.4 \times 10^{-8} \sqrt{\phi_B M_B T}}{\mu_B v_A^{0.6}}$$

with D_{AB} cm^2/s , T K, M_B g/mol, μ_B in centipoise;
 v_A , the solute molar volume at its normal B.P., cm^3/mol ;
 ϕ_B “solvent association factor” (H_2O 2.6, CH_3OH 1.9)

TABLE 3.3

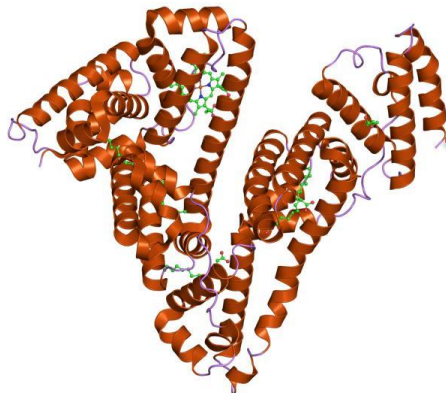
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Consider diffusivity of biologically important solutes.

The diffusion coefficients for the large protein molecules are on the order of magnitude of $5 \times 10^{-11} \text{ m}^2/\text{s}$ compared to the values of about $1 \times 10^{-9} \text{ m}^2/\text{s}$ for the small solutes ---- This means macromolecules diffuse at a rate about 20 times as slow as small solute molecules for the same concentration differences.



Diffusivities for Dilute Biological Solutes in Aqueous Solution

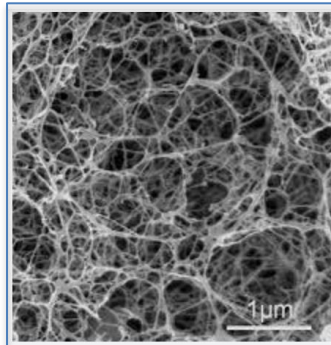
Solute	Temperature		Diffusivity (m^2/s)	Molecular Weight
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Urea	20	293	1.20×10^{-9}	60.1
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Urease	25	298	4.01×10^{-11}	482 700
	20	293	3.46×10^{-11}	
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Creatinine	37	310	1.08×10^{-9}	113.1
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Consider diffusivity of biological nutrients in gels: Diffusion through a porous structure.

At right, agar gel used in cell-growth media, chromatography, and gel electrophoresis.

If a structure is open enough, diffusivity is proportional to % porosity (void fraction ϵ), so

$$D_{effective,AB} = \epsilon \cdot D_{AB}.$$



Solute Diffusivities in Dilute Biological Gels in Aqueous Solution

Solute	Gel	Wt % Gel in Solution	Temperature		Diffusivity (m^2/s)
			K	°C	
Sucrose	Gelatin	0	278	5	0.285×10^{-9}
		3.8	278	5	0.209×10^{-9}
		10.35	278	5	0.107×10^{-9}
		5.1	293	20	0.252×10^{-9}
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		5.1	278	5	0.609×10^{-9}
		10.0	278	5	0.542×10^{-9}
		5.1	293	20	0.859×10^{-9}
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		6.02	278	5	0.199×10^{-9}
Dextrose	Agar	0.79	278	5	0.327×10^{-9}
Sucrose	Agar	0.79	278	5	0.247×10^{-9}
Ethanol	Agar	5.15	278	5	0.393×10^{-9}
NaCl (0.05 M)	Agarose	0	298	25	1.511×10^{-9}
		2	298	25	1.398×10^{-9}

Diffusion/diffusivity in gas mixtures: The Stefan-Maxwell equation.

- In gases, D_{AB} is normally assumed to be independent of composition.
- Multi-component diffusion can then be described by the Stefan-Maxwell Equation:

$$\frac{dx_i}{dz} = \sum_{j=1}^n \frac{c_i c_j}{c^2 D_{ji}} \left(\frac{J_j}{c_j} - \frac{J_i}{c_i} \right)$$

- Blanc's Law for mixture diffusivity in gases is then a useful simplification.
 - If dilute i diffuses into a homogeneous mixture, then $J_j \cong 0$, so $\frac{dx_i}{dz} = \sum_{j=1}^n \frac{c_i c_j}{c^2 D_{ji}} \left(\frac{J_j}{c_j} - \frac{J_i}{c_i} \right)$.
 - Use $\frac{c_j}{c} = x_j$ and $J_{ij} = D_{i,mix} c \frac{dx_i}{dz}$; then solve to get

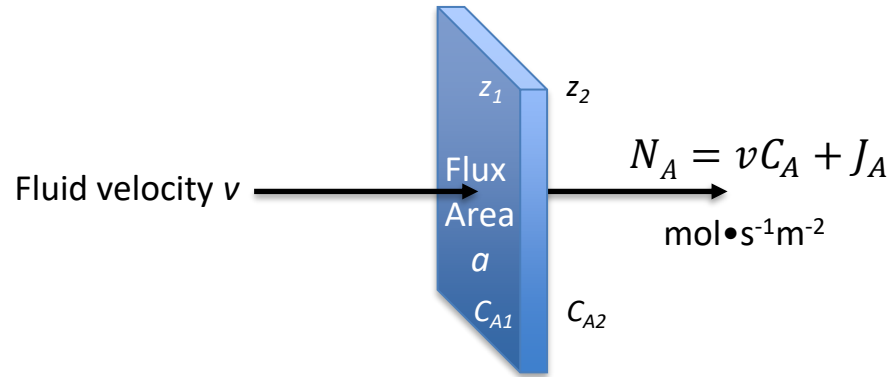
$$D_{i,mix} \equiv \frac{-J_i}{c \cdot dx_i/dz} = \left(\sum_{j=1, j \neq i}^n \frac{x_j}{D_{ij}} \right)^{-1}$$

Transition:

II.e. Differential species-continuity equations for practical use.

- Constructed from flux-based “shell balances.”
 - Solve with boundary conditions.
- Set up and examine unsteady-state diffusion problems.
- Use analytical solutions or Gurney-Lurie graphs to solve problems.

Build a basic time-dependent, 1-D shell balance:



Can include reaction: Net volumetric generation rate of A, $R_A(z_{ave}, t)$ in $(\text{mol/s})/\text{m}^3$

Accumulation rate (mol/s) = Rate in – Rate out + Net Generation

$$\frac{\partial \{ (C_A(z_{ave})) \cdot (z_2 - z_1) a \}}{\partial t} = (N_A a)_1 - (N_A a)_2 + R_A(z_{ave}) \cdot (z_2 - z_1) a$$

$$\frac{\partial C_A(z_{ave}, t)}{\partial t} + \frac{(N_A)_2 - (N_A)_1}{z_2 - z_1} = R_A(z_{ave}), \quad \text{and as } \Delta z \rightarrow dz, \quad \frac{\partial C_A(z, t)}{\partial t} + \frac{\partial N_A}{\partial z} = R_A(z)$$

$$\frac{\partial C_A}{\partial t} + \frac{\partial v C_A}{\partial z} = -\frac{\partial J_A}{\partial z} + R_A = \frac{\partial}{\partial z} \left(D_{AB} \frac{\partial C_A}{\partial z} \right) + R_A; \quad \text{can become} \quad \frac{\partial C_A}{\partial t} + v \frac{\partial C_A}{\partial z} = D_{AB} \frac{\partial^2 C_A}{\partial z^2} + R_A$$

Expanded species-continuity equations in rectilinear, cylindrical, & spherical coordinates.

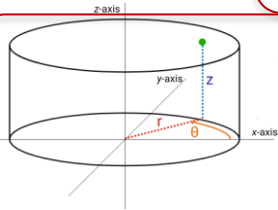
TABLE 18.2-1

THE EQUATION OF CONTINUITY OF A IN VARIOUS COORDINATE SYSTEMS

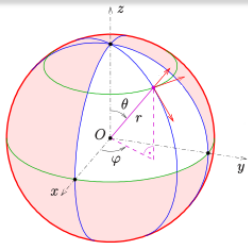
(Eq. 18.1-10)

Rectangular coordinates:

$$\frac{\partial c_A}{\partial t} + \left(\frac{\partial N_{Ax}}{\partial x} + \frac{\partial N_{Ay}}{\partial y} + \frac{\partial N_{Az}}{\partial z} \right) = R_A \quad (A)$$

*Cylindrical coordinates:*

$$\frac{\partial c_A}{\partial t} + \left(\frac{1}{r} \frac{\partial}{\partial r} (r N_{Ar}) + \frac{1}{r} \frac{\partial N_{A\theta}}{\partial \theta} + \frac{\partial N_{Az}}{\partial z} \right) = R_A \quad (B)$$

*Spherical coordinates:*

$$\frac{\partial c_A}{\partial t} + \left(\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 N_{Ar}) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (N_{A\theta} \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial N_{A\phi}}{\partial \phi} \right) = R_A \quad (C)$$

Expanded equations: Constant density and Fickian diffusion in rectilinear, cylindrical, & spherical coordinates.

TABLE 18.2-2

THE EQUATION OF CONTINUITY OF A FOR CONSTANT ρ AND $\mathcal{D}_{AB}$
(Eq. 18.1-17)

Rectangular coordinates:

$$\frac{\partial c_A}{\partial t} + \left(v_x \frac{\partial c_A}{\partial x} + v_y \frac{\partial c_A}{\partial y} + v_z \frac{\partial c_A}{\partial z} \right) = \mathcal{D}_{AB} \left(\frac{\partial^2 c_A}{\partial x^2} + \frac{\partial^2 c_A}{\partial y^2} + \frac{\partial^2 c_A}{\partial z^2} \right) + R_A \quad (A)$$

Cylindrical coordinates:

$$\begin{aligned} \frac{\partial c_A}{\partial t} + \left(v_r \frac{\partial c_A}{\partial r} + v_\theta \frac{1}{r} \frac{\partial c_A}{\partial \theta} + v_z \frac{\partial c_A}{\partial z} \right) \\ = \mathcal{D}_{AB} \left(\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial c_A}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 c_A}{\partial \theta^2} + \frac{\partial^2 c_A}{\partial z^2} \right) + R_A \end{aligned} \quad (B)$$

Spherical coordinates:

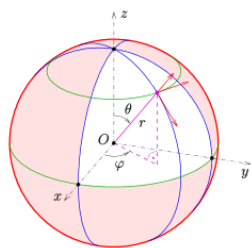
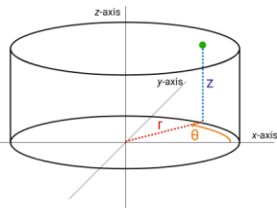
$$\begin{aligned} \frac{\partial c_A}{\partial t} + \left(v_r \frac{\partial c_A}{\partial r} + v_\theta \frac{1}{r} \frac{\partial c_A}{\partial \theta} + v_\phi \frac{1}{r \sin \theta} \frac{\partial c_A}{\partial \phi} \right) \\ = \mathcal{D}_{AB} \left(\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_A}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial c_A}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 c_A}{\partial \phi^2} \right) + R_A \end{aligned} \quad (C)$$

Or more concisely & generally,

$$\frac{\partial C_A}{\partial t} + \vec{\nabla} \cdot \vec{v} C_A = \vec{\nabla} \cdot D_{AB} \vec{\nabla} C_A + R_A$$

where for constant C and D_{AB} ,

$$\frac{\partial C_A}{\partial t} + \vec{v} \cdot \vec{\nabla} C_A = D_{AB} \vec{\nabla}^2 C_A + R_A$$



What can we do with these equations?

- De-dimensionalize them to reveal the governing dimensionless parameters and terms.
- Solve and apply them to answer significant engineering questions.

$$v_z \frac{\partial c_A}{\partial z} = D_{AB} \frac{\partial^2 c_A}{\partial z^2} - k c_A$$

BC (Initial condition): $c_A = c_{A0}$ at $z=0$. [Would need a second BC to solve.]

Choose dimensionless variables:

$$Z \equiv \frac{z}{L} \text{ where } L \text{ is reactor length, so } z = L \cdot Z; C \equiv \frac{c_A}{c_{A0}}, \text{ so } c_A = c_{A0} \cdot C$$

De-dimensionalize:

$$\frac{v_z c_{A0}}{L} \frac{\partial C}{\partial Z} = \frac{D_{AB} c_{A0}}{L^2} \frac{\partial^2 C}{\partial Z^2} - (k c_{A0}) C$$

$$\left(\frac{v_z L}{D_{AB}} \right) \frac{\partial C}{\partial Z} = \frac{\partial^2 C}{\partial Z^2} - \left(\frac{k L^2}{D_{AB}} \right) C; \quad C = 1 \text{ at } Z=0$$

$$\begin{aligned} \frac{\partial c_A}{\partial z} &= \frac{\partial (c_{A0} \cdot C)}{\partial (L \cdot Z)} = \frac{c_{A0}}{L} \frac{\partial C}{\partial Z} \\ \frac{\partial}{\partial z} \frac{\partial c_A}{\partial z} &= \frac{\partial}{\partial (L \cdot Z)} \frac{\partial (c_{A0} \cdot C)}{\partial (L \cdot Z)} = \frac{c_{A0}}{L^2} \frac{\partial^2 C}{\partial Z^2} \end{aligned}$$

FYI: $\frac{v_z L}{D_{AB}}$ is a Peclet number;

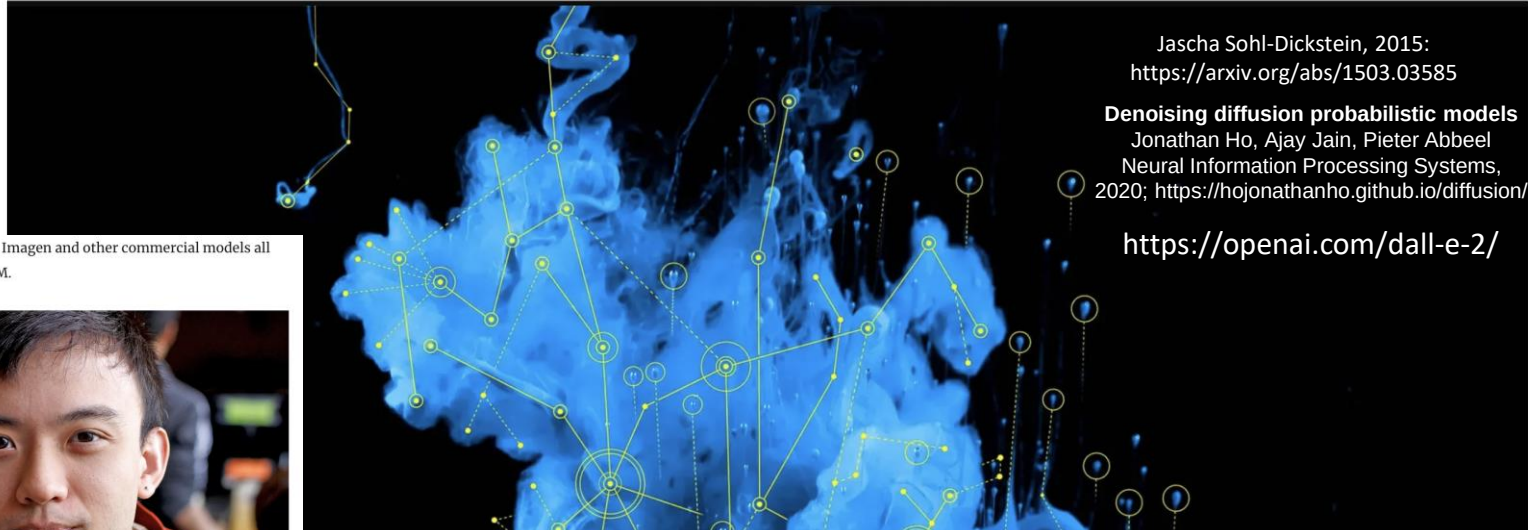
$\frac{k L^2}{D_{AB}}$ is a Damkohler number

Use these equations for time-dependent mass transfer.

- An important set of cases: “Transient” or “Unsteady-state” mass transfer.
 - How long to dry a product?
 - Release time of an API (active pharmaceutical ingredient) from a pill?
 - Generative AI for images and text...

The Physics Principle That Inspired Modern AI Art

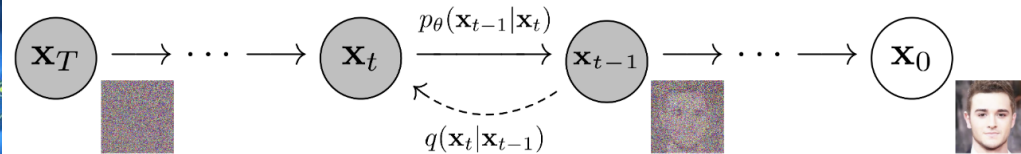
Diffusion models generate incredible images by learning to reverse the process that, among other things, causes ink to spread through water.



DALL·E 2, Stable Diffusion, Imagen and other commercial models all use some variation of DDPM.



Jonathan Ho and his colleagues combined Sohl-Dickstein and Song's methods to make possible modern diffusion models, such as DALL·E 2.



Diffusion probabilistic models are parameterized Markov chains trained to gradually denoise data. We estimate parameters of the generative process p .

Consider unsteady-state mass transfer.

Previously, we did pseudo-SS approach (like rate-of-evaporation problems);

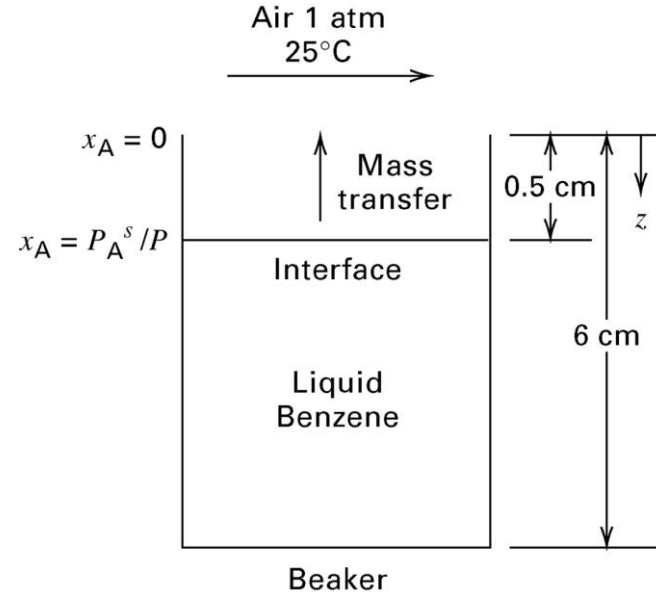
Now time-dependent $C(\mathbf{x}, t)$ due to diffusion.

- (1) Simplify and solve differential equation with boundary conditions.
- (2) Most generally with dimensionless C as dimensionless position, dimensionless time $D_{AB}t/a^2$, where a is characteristic length, and dimensionless parameters.
- (3) For combined geometries (cube, finite cylinder), multiply dimensionless solutions.

For certain cases, we already have analytic or graphical solutions we can use.

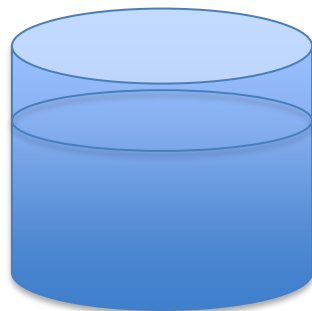
First, let's review the pseudo-SS diffusion problem...

- Consider a 6.0-cm-deep beaker, filled with liquid benzene to 0.50 cm from the top.
- Dry air at 25°C and 1.000 atm is blown gently across the top, keeping the benzene vapor concentration there at ~zero.
- What is the initial rate of evaporation?
- How long until benzene drops 2 cm?



Approach to the problem:

- Sketch.



$$P_{\text{vapA}}(25^\circ\text{C}) = 0.131 \text{ atm}$$

Interface area a

$$\rho_A = 0.874 \text{ g/cm}^3$$

- Identify or develop equations:
 - Benzene vapor (A) diffuses through stagnant air.

$$N_A = \frac{\hat{e} D_{AB} C}{\hat{e} (z - z_1) (x_B)_{\log mean}} \frac{\hat{u}}{\hat{u}} (x_{A1} - x_A)$$

What is the initial rate of evaporation?

$$N_A = \frac{\hat{e} D_{AB} C}{\hat{e} (z - z_1) (x_B)_{\log mean}} \hat{u} (x_{A1} - x_A)$$

Obtain $N_{A,0}$:

- If $z=0$ is the lip, then $(z-z_1)=-0.50$ cm
- $x_{A1}-x_{A2} = 0-P_{\text{vapA}} = -0.131$ atm
- $x_{B,\log mean} = (1-0.869)/\ln(1/.869) = 0.933$
- $C=P/RT = 1.000/(82.06 \times 298) = 4.09 \times 10^{-5}$ mol/cm³
- $D_{AB}(25^\circ\text{C}, 1 \text{ atm}) = 0.0905$ cm²/s
- $N_{A,0} = 1.04 \times 10^{-6}$ mol·s⁻¹cm⁻²

How long until benzene drops 2 cm?

- Write balance at the interface:

$$N_A \cdot a \cdot dt = (\rho/MW)_{liq} \cdot a \cdot dz$$

- Integrate:

$$\int_{z=0}^z \frac{D_{AB} C}{(x_B)_{\log mean}} (1 - x_A) dz = \frac{r}{MW} a \int_0^t dt$$

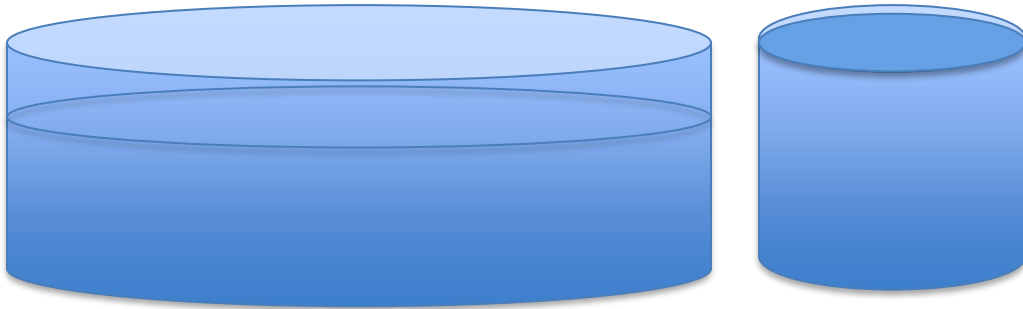
$$\int_{z_1}^{z_2} \frac{r}{MW} \frac{(x_B)_{\log mean}}{D_{AB} C x_A} dz = \int_0^t dt$$

$$t = \frac{r}{MW} \frac{(x_B)_{\log mean}}{D_{AB} C x_A} \times \frac{1}{2} (z_2^2 - z_1^2)$$

$$= \frac{0.874}{78.11} \frac{0.939}{(0.0905)(4.09 \times 10^{-5})(0.131)} \times \frac{1}{2} (2.5^2 - 0.5^2) = 17.9 \text{ hr}$$

Consider the “What Ifs”:

- What if there is a T effect from evaporative cooling?
- What if the container is much narrower?
- What if air is blown faster across the top?
- What if the container is wider?
- What if it starts brimful?



$$\dot{Q}_0^t = \frac{r}{MW} \frac{(x_B)_{\log mean}}{D_{AB} C x_A} \dot{Q}_{z_1}^{z_2} dz$$

$$t = \frac{r}{MW} \frac{(x_B)_{\log mean}}{D_{AB} C x_A} \times \frac{1}{2} \left(z_2^2 - z_1^2 \right)$$

For true time dependence, three choices on how to proceed:

- Easiest: Plug into a pre-solved equation (an analytic solution).
- If not, get quick estimates by using unsteady-state diffusion graphs.
 - Developed by Gurney and Lurie (1923) and Heisler (1947) from solutions to the differential equations – and still useful.
 - Figures provided here.
- Develop your own numerical solution from the differential equations.
 - Set up the differential equation and BCs; de-dimensionalize & solve.
 - MATLAB has a function for solving partial differential equations.
 - Can also use simulation codes like COMSOL.

Easiest is to use an already-solved equation.

- Solve analytically or by numerical methods
- For drying a slab, there is an analytical solution

for $Y \equiv \frac{(c_{A1}-c_A)}{(c_{A1}-c_{A0})}$, a Fourier series in $\theta \equiv \frac{D_{AB}t}{x_1^2}$ and $X \equiv \frac{x}{x_1}$:

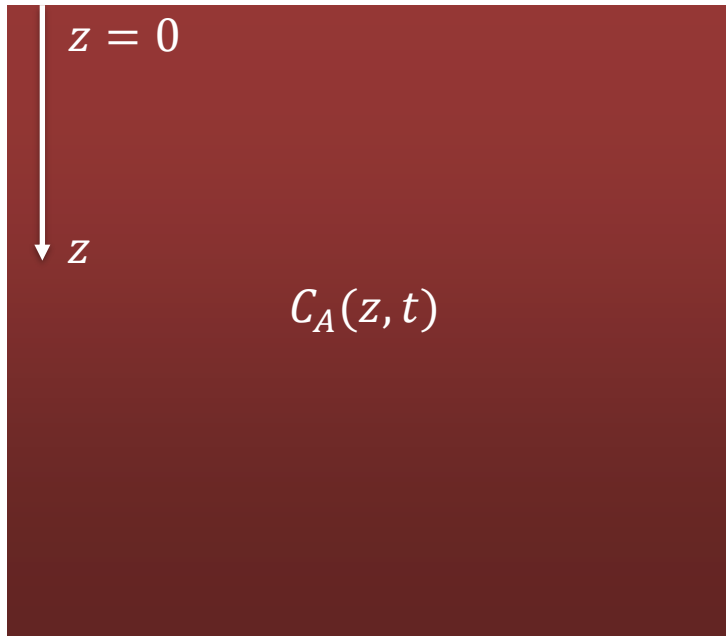
$$Y = \frac{4}{\pi} \left[\frac{1}{1} \exp\left(-\frac{1^2\pi^2\theta}{4}\right) \sin\left(\frac{1\pi X}{2}\right) + \frac{1}{3} \exp\left(-\frac{3^2\pi^2\theta}{4}\right) \sin\left(\frac{3\pi X}{2}\right) + \frac{1}{5} \exp\left(-\frac{5^2\pi^2\theta}{4}\right) \sin\left(\frac{5\pi X}{2}\right) + \dots \right]$$



- Equivalent to equation for diffusion into a flat plate.
- If you can obtain such an analytical solution, it can be used for results and to help validate numerical solutions.

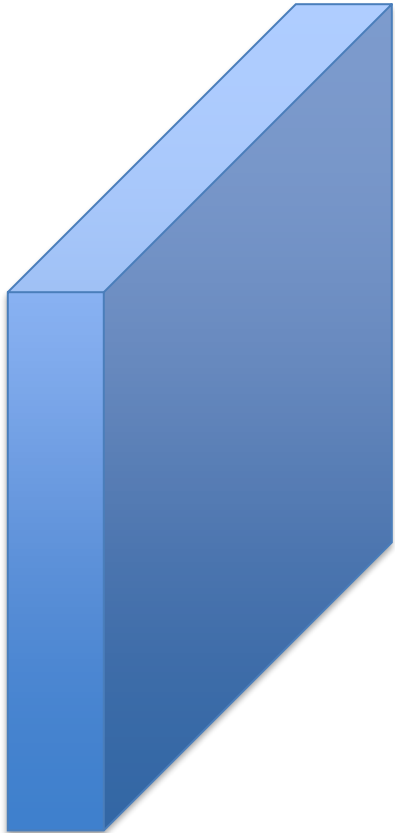
Consider diffusion in a semi-infinite (stagnant) medium.

How deeply will a spill of trichloroethylene penetrate into clay in 1 d?



- Use 1-D pure-diffusion case $\frac{\partial C_A}{\partial t} = D_{AB} \frac{\partial^2 C_A}{\partial z^2}$ (“Fick’s Second Law”) with prior ground concentration $C_{A0} = 0$ and imposed surface concentration C_{As} .
- Expressing it as “fractional concentration change” $\theta = \frac{C_A - C_{A0}}{C_{As} - C_{A0}}$, a “deviation variable,” gives $\theta = \text{erfc}\left(\frac{z}{2\sqrt{D_{AB}t}}\right)$ where $\text{erfc}(x) = 1 - \frac{1}{\sqrt{\pi}} \int_0^x e^{-\eta^2} d\eta$
- Define penetration depth $z_p \equiv 4\sqrt{D_{AB}t}$ where $\theta = 0.0047$ or 0.5% of C_{As} .
- Using that and $D_{AB} = 4 \cdot 10^{-10} \text{ m}^2/\text{s}$, $z_p = 0.02 \text{ m} = 2 \text{ cm}$.

For certain cases, we can use pre-solved graphical solutions.

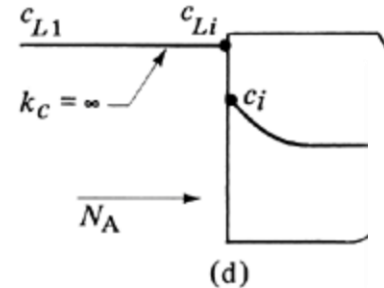
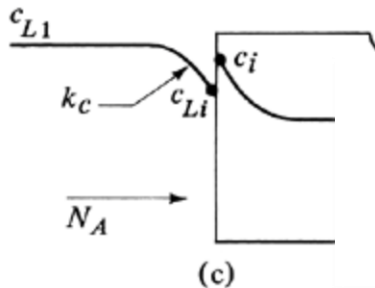
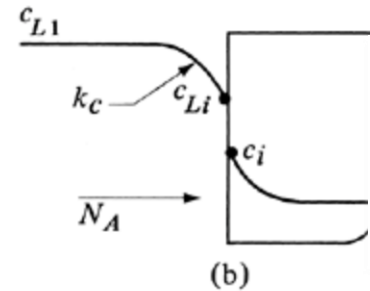
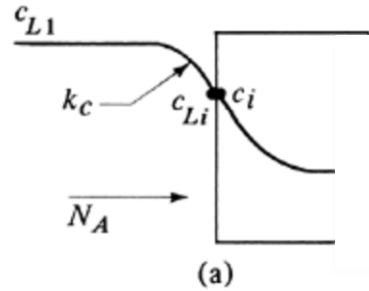


- Dry a 2.00-inch-thick slab, initially being 30.% H₂O by weight. It is dried by blowing dry air past it, where the air contains 5.0 wt % H₂O and the convective mass transfer coefficient of water into the blowing air is high.
 - The slab is much longer and wider than 2 inches, so the effects of drying through the edges are negligible.
 - Further suppose the drying is by diffusion with a diffusivity of $3.72 \cdot 10^{-6}$ m²/hr.
- **How long will it take to dry the wood to 10% H₂O at the center?**
- At center, $X \equiv \frac{x}{x_1} = 0$; but $Y(0)$ and $Y(1) = \frac{(c_{A1} - c_A)}{(c_{A0} - c_{A1})}$ where we need a value of c_{A1} to proceed: 0? 5%?

If the surface boundary condition is phase equilibrium, use Equilibrium Distribution Coefficient K_i ; otherwise, a mass-transfer coefficient k .

Interface conditions for convective mass transfer and an equilibrium distribution coefficient $K = c_{Li}/c_i$: (a) $K = 1$.

(b) $K > 1$, (c) $K < 1$, (d) $K > 1$ and $k_c = \infty$.



Relation Between Mass- and Heat-Transfer Parameters for Unsteady-State Diffusion^[*]

Find the right graph and variables.

- Examine terms.
- Identify appropriate dimensionless parameters for analyzing unsteady-state transport.
- For our slab, use flat plate model.

If equilibrium at surface,

$$k_c A (c_{air} - c_{air,s}) = -D_{AB} A \frac{\partial c}{\partial x}$$

Heat Transfer	Mass Transfer	
	$K = c_L/c = 1.0$	$K = c_L/c \neq 1.0$
$Y, \frac{T_1 - T}{T_1 - T_0}$ $1 - Y, \frac{T - T_0}{T_1 - T_0}$	$\frac{c_1 - c}{c_1 - c_0} = Y$ $\frac{c - c_0}{c_1 - c_0}$	$\frac{c_1/K - c}{c_1/K - c_0}$ $\frac{c - c_0}{c_1/K - c_0}$
$X, \frac{\alpha t}{x_1^2}$ $\frac{x}{2\sqrt{\alpha t}}$	$\frac{D_{AB} t}{x_1^2} = \theta$ $\frac{x}{2\sqrt{D_{AB} t}}$	$\frac{D_{AB} t}{x_1^2}$ $\frac{x}{2\sqrt{D_{AB} t}}$
$m, \frac{k}{hx_1}$ $\frac{h}{k}\sqrt{\alpha t}$	$\frac{D_{AB}}{k_c x_1}$ $\frac{k_c}{D_{AB}}\sqrt{D_{AB} t}$	$\frac{D_{AB}}{K k_c x_1}$ $\frac{K k_c}{D_{AB}}\sqrt{D_{AB} t}$
$n, \frac{x}{x_1}$	$\frac{x}{x_1} = X$	$\frac{x}{x_1}$

[*] x is the distance from the center of the slab, cylinder, or sphere; for a semi-infinite slab, x is the distance from the surface, c_0 is the original uniform concentration in the solid, c_1 the concentration in the fluid outside the slab, and c the concentration in the solid at position x and time t .

Find the right graph and variables.

- For the slab,
 $L=2.00 \text{ in} \Rightarrow x_1=1.00 \text{ in.}=0.0254 \text{ m};$
- $\Rightarrow n = x/x_1 = 0$ (center).
- Bulk concn $c_L=0.050$; assume $K=1.0$ (maybe?).
 At time t , $Y = (c_1 - c(t))/(c_1 - c_0)$
 $= (0.050 - 0.10)/(0.050 - 0.30) = 0.20$.
- The convective mass transfer coefficient k_c is high, so $m = \frac{D_{AB}}{k_c x_1} \cong 0$ So $Y=0.20$,
 $n=0.0$, and $m=0$.

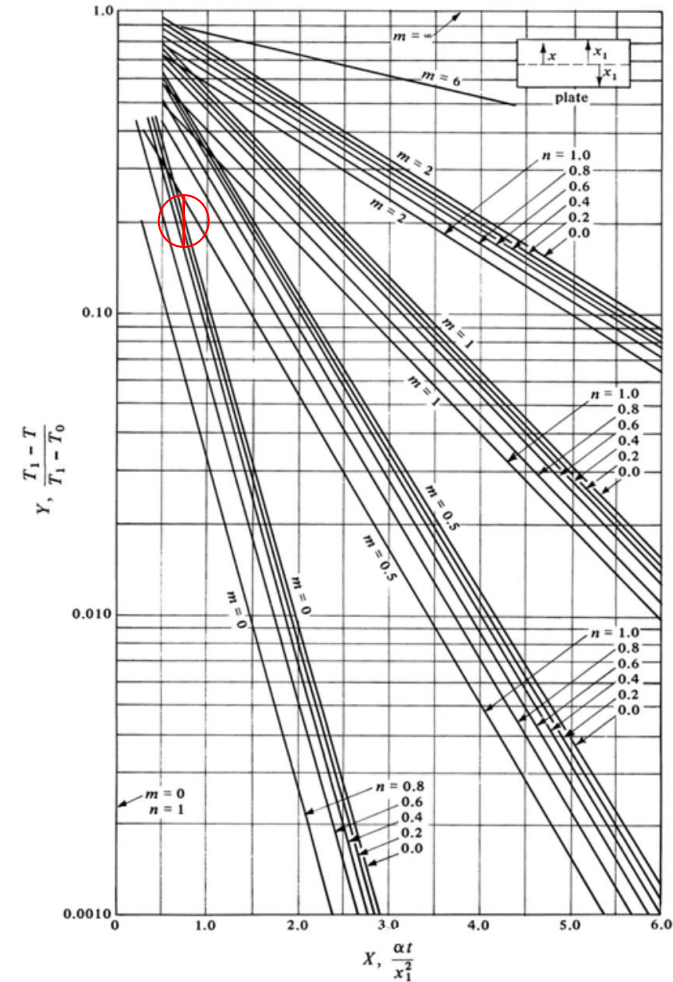
Relation Between Mass- and Heat-Transfer Parameters for Unsteady-State Diffusion^[*]

Heat Transfer	Mass Transfer	
	$K = c_L/c = 1.0$	$K = c_L/c \neq 1.0$
$Y, \frac{T_1 - T}{T_1 - T_0}$	$\frac{c_1 - c}{c_1 - c_0}$	$\frac{c_1/K - c}{c_1/K - c_0}$
$1 - Y, \frac{T - T_0}{T_1 - T_0}$	$\frac{c - c_0}{c_1 - c_0}$	$\frac{c - c_0}{c_1/K - c_0}$
$X, \frac{\alpha t}{x_1^2}$	$\frac{D_{AB} t}{x_1^2}$	$\frac{D_{AB} t}{x_1^2}$
$\frac{x}{2\sqrt{\alpha t}}$	$\frac{x}{2\sqrt{D_{AB} t}}$	$\frac{x}{2\sqrt{D_{AB} t}}$
$m, \frac{k}{hx_1}$	$\frac{D_{AB}}{k_c x_1}$	$\frac{D_{AB}}{K k_c x_1}$
$\frac{h}{k} \sqrt{\alpha t}$	$\frac{k_c}{D_{AB}} \sqrt{D_{AB} t}$	$\frac{K k_c}{D_{AB}} \sqrt{D_{AB} t}$
$n, \frac{x}{x_1}$	$\frac{x}{x_1}$	$\frac{x}{x_1}$

^[*] x is the distance from the center of the slab, cylinder, or sphere; for a semi-infinite slab, x is the distance from the surface, c_0 is the original uniform concentration in the solid, c_1 the concentration in the fluid outside the slab, and c the concentration in the solid at position x and time t .

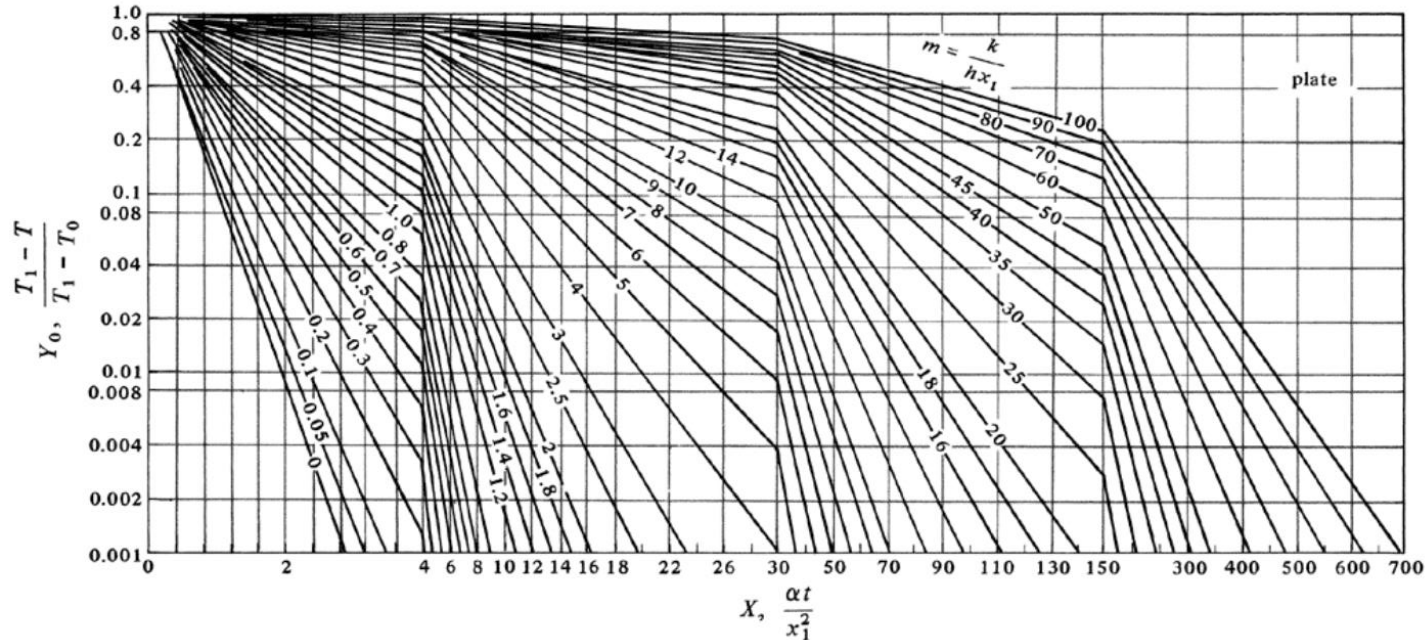
Find the right graph and variables.

- For the plank,
 $L=2.00 \text{ in} \Rightarrow x_1=1.00 \text{ in.}=0.0254 \text{ m}$;
- $\Rightarrow n = x/x_1 = 0$ (center).
- Bulk concn $c_L=0.050$; assume $K=1.0$ (maybe?).
 At time t , $Y = (c_1 - c(t))/(c_1 - c_0)$
 $= (0.050 - 0.10)/(0.050 - 0.30) = 0.20$.
- The convective mass transfer coefficient k_c is high, so $m = \frac{D_{AB}}{k_c x_1} \cong 0$ So $Y=0.20$, $n=0.0$, and $m=0$.
- Figure gives $X = D_{AB}t/x_1^2 \sim 0.75$.
- $D_{AB}=3.72 \cdot 10^{-6} \text{ m}^2/\text{hr}$, so $t=130. \text{ hr}$



Alternatively, use figure for center value.

Chart for determining temperature at the center of a large flat plate for unsteady-state heat conduction. [From H. P. Heisler, Trans. A.S.M.E., 69, 227 (1947). With permission.]



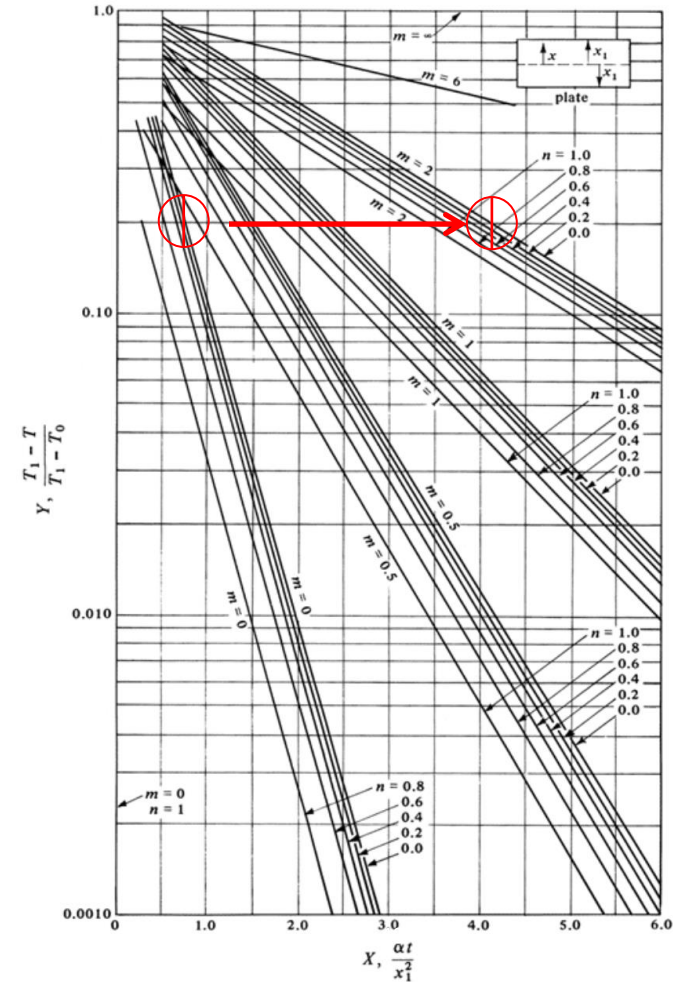
At $Y=0.20$ and $m=0$, figure ($n=0.0$) gives $X \sim 0.70 = D_{AB}t/x_1^2$. Then $t=128$. hr

Consider complications:

- How does the result change if the air flow stops and convective mass-transfer term $k_c L$ in m decreases to $1.86 \cdot 10^{-6} \text{ m}^2/\text{hr}$? ($L=x_1$)
- Repeat the analysis of (a) with $m=D_{AB}/k_c x_1 = (3.72 \cdot 10^{-6})/(1.86 \cdot 10^{-6})$, so $m=2.00$.
- $Y=0.20$ and $n=0.0$ still, so $X=4.15$.

Then $t=Xx_1^2/D_{AB}=720 \text{ hr}$ (vs 130 hr before).

Unsteady-state heat conduction in a large flat plate. [From H. P. Gurney and J. Lurie, Ind. Eng. Chem., 15, 1170 (1923).]



Similarly, you can do cylinders.

Unsteady-state heat conduction in a long cylinder. [From H. P. Gurney and J. Lurie, Ind. Eng. Chem., 15, 1170 (1923).]

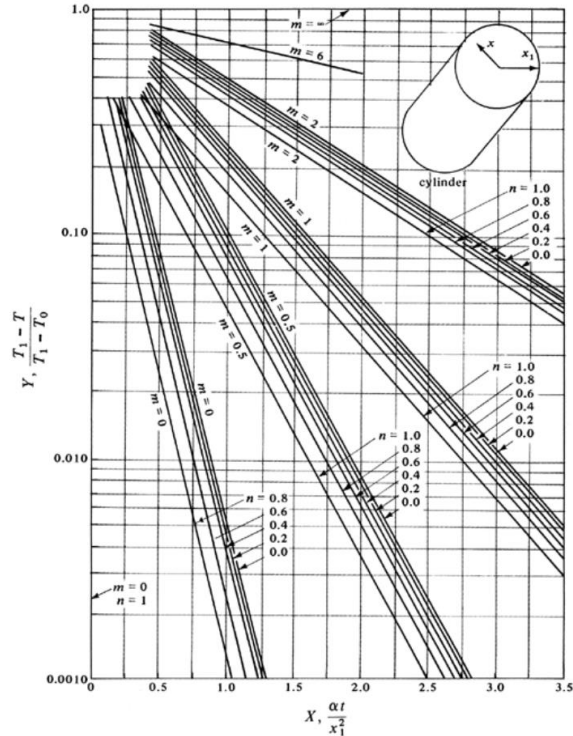
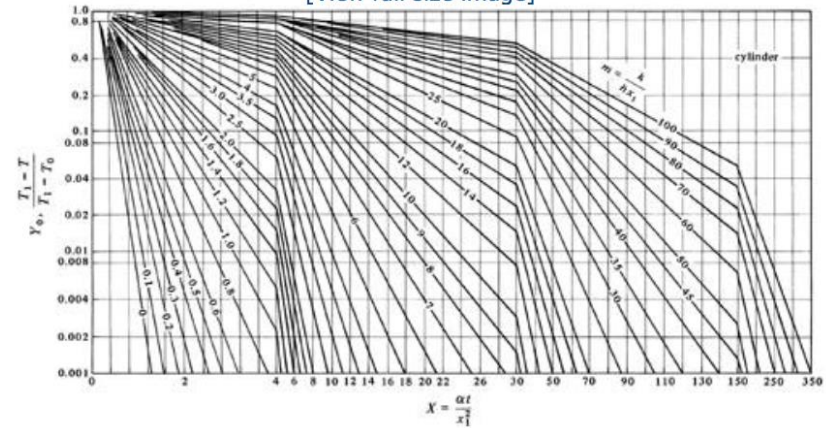


Chart for determining temperature at the center of a long cylinder for unsteady-state heat conduction. [From H. P. Heisler, Trans. A.S.M.E., 69, 227 (1947). With permission.]

[View full size image]

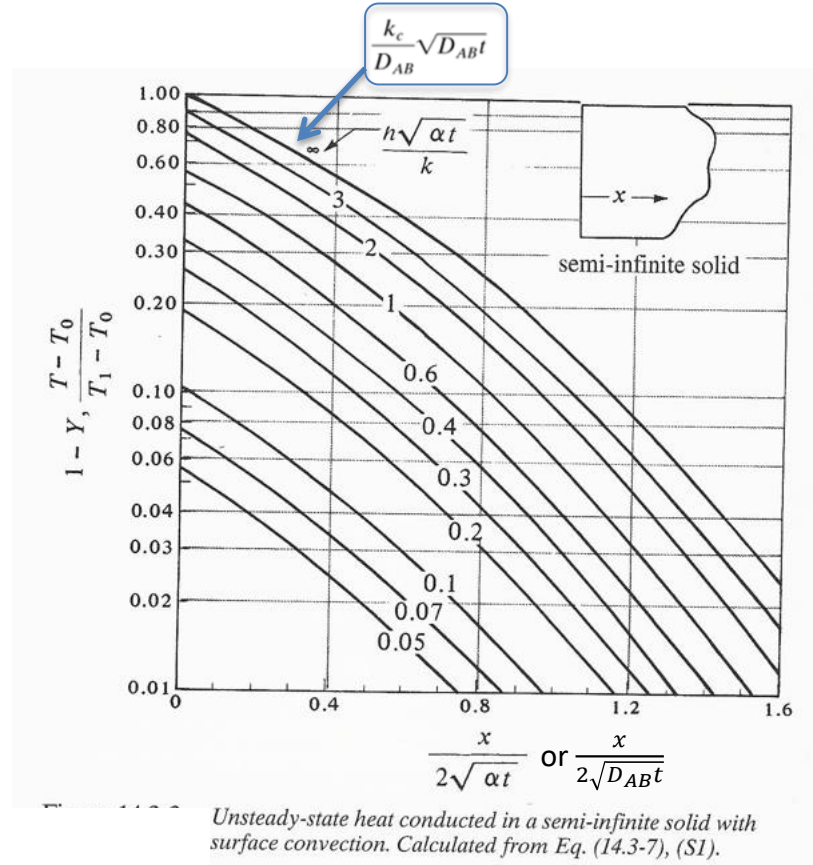


Diffusion into/from a semi-infinite slab is a little different.

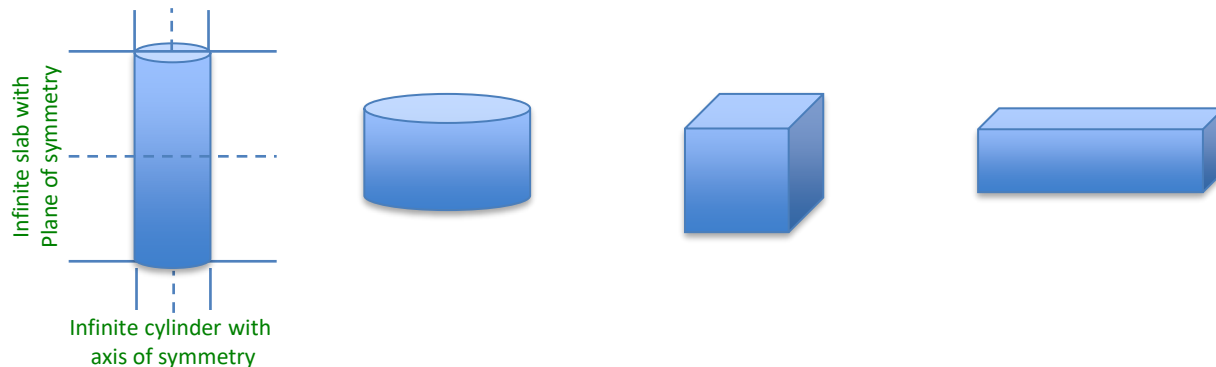
- There is no reference x ,
- But the equations give dimensionless form as

$$\frac{x}{2\sqrt{D_{AB}t}}$$

- And for surface mass transfer, $\frac{k_c}{D_{AB}}\sqrt{D_{AB}t}$



What if you have a different geometry?



- Multiply the **dimensionless solutions**.
 - For the first two, $Y_{cylinder}(n_{cyl}, X) \cdot Y_{slab}(n_{slab}, X)$
 - Or the next two, $Y_{slab, x}(n_{slab, x}, X) \cdot Y_{slab, y}(n_{slab, y}, X) \cdot Y_{slab, z}(n_{slab, z}, X)$
 - Because diffusivity is in $X = Dt/x_1^2$, you can even use a different diffusivity in each direction if that is the case.

Consider an example for pharma mfg:

- A spherical pill, 6.0 mm diameter, contains an active ingredient that diffuses out when the pill becomes wet. The pill is made by mixing the dose of 25.0 mg with an inactive, non-dissolving powder (that makes up most of the pill) and compacting it into a semi-porous, spherical pill, $V=(4/3)\pi r^3$. Assume the active ingredient is mixed very uniformly.
- The active ingredient's effective diffusivity through the pill is determined by putting the pill in a container of stirred, clean water held at 37°C, removing the pill after a specified time period, and analyzing the concentration of active ingredient remaining at the center: 55 mg/cm³ after 80 min. Experimentally, the pill size didn't change; assume negligible external mass-transfer resistance.
- **For design of optimal drug release, find the diffusivity.**

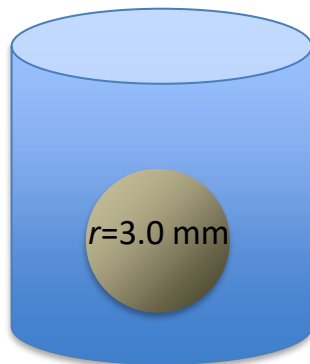
Approach to the problem:

- Sketch.

$$C_{\text{center},0} = 25 \text{ mg}/[1.33\pi(0.30\text{cm})^3] \\ = 220 \text{ mg/cm}^3$$

$$C_{\text{center},80\text{min}} = 55 \text{ mg/cm}^3$$

$$C_{\text{surface}} \sim 0 \text{ for all time}$$



- Compute Y, m, n ; find X .
- Identify which graph to use:
 - Un-SS conduction in a sphere.

Relation Between Mass- and Heat-Transfer Parameters for Unsteady-State Diffusion^[*]

Heat Transfer	Mass Transfer	
	$K = c_L/c = 1.0$	$K = c_L/c \neq 1.0$
$Y, \frac{T_1 - T}{T_1 - T_0}$	$\frac{c_1 - c}{c_1 - c_0}$	$\frac{c_1/K - c}{c_1/K - c_0}$
$1 - Y, \frac{T - T_0}{T_1 - T_0}$	$\frac{c - c_0}{c_1 - c_0}$	$\frac{c - c_0}{c_1/K - c_0}$
$X, \frac{\alpha t}{x_1^2}$	$\frac{D_{AB}t}{x_1^2}$	$\frac{D_{AB}t}{x_1^2}$
$\frac{x}{2\sqrt{\alpha t}}$	$\frac{x}{2\sqrt{D_{AB}t}}$	$\frac{x}{2\sqrt{D_{AB}t}}$
$m, \frac{k}{hx_1}$	$\frac{D_{AB}}{Kk_c x_1}$	$\frac{D_{AB}}{Kk_c x_1}$
$\frac{h}{k}\sqrt{\alpha t}$	$\frac{k_c}{D_{AB}}\sqrt{D_{AB}t}$	$\frac{Kk_c}{D_{AB}}\sqrt{D_{AB}t}$
$n, \frac{x}{x_1}$	$\frac{x}{x_1}$	$\frac{x}{x_1}$

^[*] x is the distance from the center of the slab, cylinder, or sphere; for a semi-infinite slab, x is the distance from the surface, c_0 is the original uniform concentration in the solid, c_1 the concentration in the fluid outside the slab, and c the concentration in the solid at position x and time t .

Apply parameters:

$$x=0; x_1=3.0\text{mm}$$

$$K=C_L/C_{surf}=1.0 (?)$$

$$m=D_{AB}/k_c x_1 \rightarrow 0$$

$$n=x/x_1=0$$

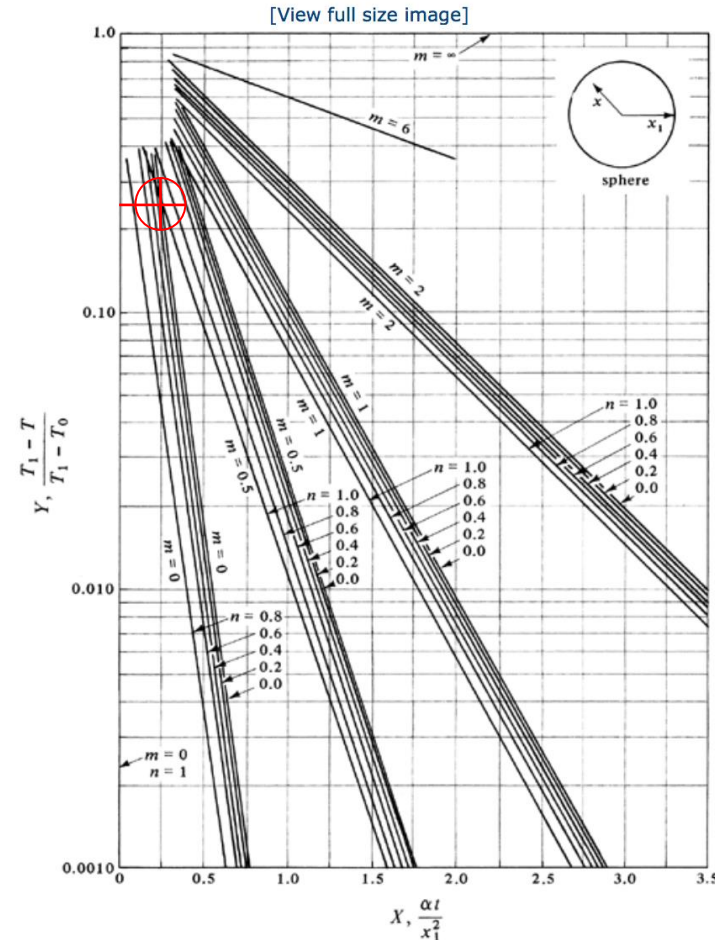
$$Y=(C_1-C)/(C_1-C_0)=(0-55)/(0-220)=0.25$$

From graph, read

$$X=D_{AB}t/x_1^2=0.26$$

or

$$D_{AB}=0.26x_1^2/t=4.9 \times 10^{-4} \text{ cm}^2/\text{s}$$



Contrast to truly unsteady-state diffusion: Concentration gradients inside the object. An example:

- A spherical pill, 6.0 mm diameter, contains an active ingredient that diffuses out when the pill becomes wet. The pill is made by mixing the dose of 25.0 mg with an inactive, non-dissolving powder (that makes up most of the pill) and compacting it into a semi-porous, spherical pill, $V=(4/3)\pi r^3$. Assume the active ingredient is mixed very uniformly.
- The active ingredient's effective diffusivity through the pill is determined by putting the pill in a container of stirred, clean water held at 37°C, removing the pill after a specified time period, and analyzing the concentration of active ingredient remaining at the center: 55 mg/cm³ after 80 min. Experimentally, the pill size didn't change; assume negligible external mass-transfer resistance.
- **For design of optimal drug release, infer the diffusivity. Start with graphical solutions.**

How to use the species continuity equations?

- More commonly, analyze numerically with computational fluid dynamics.
- De-dimensionalizing helps reveal the appropriate dimensionless variables.
- Develop differential equations for unsteady-state diffusion in a slab & sphere:

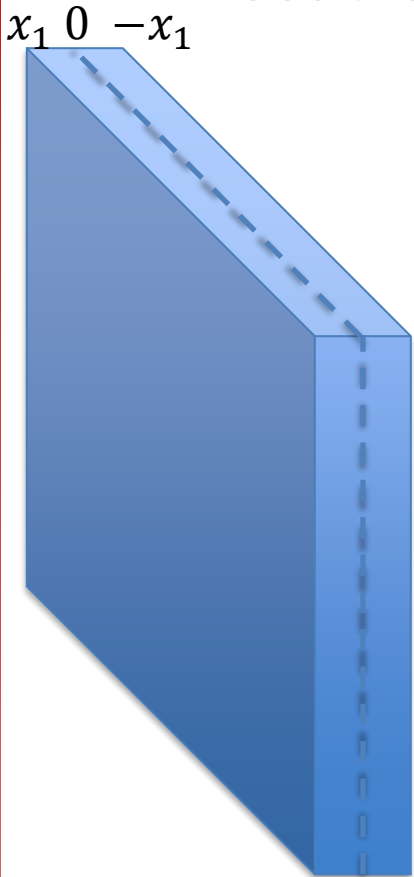
Rectangular coordinates:

$$\frac{\partial c_A}{\partial t} + \left(v_x \frac{\partial c_A}{\partial x} + v_y \frac{\partial c_A}{\partial y} + v_z \frac{\partial c_A}{\partial z} \right) = \mathcal{D}_{AB} \left(\frac{\partial^2 c_A}{\partial x^2} + \frac{\partial^2 c_A}{\partial y^2} + \frac{\partial^2 c_A}{\partial z^2} \right) + R_A$$

Spherical coordinates:

$$\begin{aligned} \frac{\partial c_A}{\partial t} + \left(v_r \frac{\partial c_A}{\partial r} + v_\theta \frac{1}{r} \frac{\partial c_A}{\partial \theta} + v_\phi \frac{1}{r \sin \theta} \frac{\partial c_A}{\partial \phi} \right) \\ = \mathcal{D}_{AB} \left(\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_A}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial c_A}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 c_A}{\partial \phi^2} \right) + R_A \end{aligned}$$

Use the species continuity equations for drying a slab.



- This equation assumes constant density and diffusivity.

Rectangular coordinates:

$$\frac{\partial c_A}{\partial t} + \left(\cancel{v_x \frac{\partial c_A}{\partial x}} + \cancel{v_y \frac{\partial c_A}{\partial y}} + \cancel{v_z \frac{\partial c_A}{\partial z}} \right) = \mathcal{D}_{AB} \left(\frac{\partial^2 c_A}{\partial x^2} + \cancel{\frac{\partial^2 c_A}{\partial y^2}} + \cancel{\frac{\partial^2 c_A}{\partial z^2}} \right) + \cancel{R_A}$$

- Infinite slab of thickness $2x_1$, where x_1 is the half-thickness.
 - Why? Assume and use symmetry: x_1 is the distance from the plane of symmetry to the surface, so $c_A(x) = c_A(-x)$.
- Account for (1) no y or z dependence;
 - (2) Solid or no-convection fluid;
 - (3) No reaction.
- To leave the governing equation: $\frac{\partial c_A}{\partial t} = D_{AB} \frac{\partial^2 c_A}{\partial x^2}$

Need boundary conditions for $\frac{\partial c_A}{\partial t} = D_{AB} \frac{\partial^2 c_A}{\partial x^2}$

x_1 0 $-x_1$

- One for $c_A(t, x)$ and two for $\frac{\partial c_A}{\partial x}(t, x)$
- For t , usually an initial condition:
 $c_A(t, x) = c_{A0}$ for $t = 0^-$ and $x_1 \geq x \geq -x_1$ (i.e., “all x ”)
- For x , usually surface condition and symmetry:
 $c_A(t, x) = c_{A1}$ for $x = x_1$ (i.e., $\pm x_1$) and $t \geq 0$ (i.e., “all t ”)
 [or possibly $c_A(t, x)$ is a convective mass-transfer function;
 more explanations later]

$$D_{AB} \frac{\partial^2 c_A}{\partial x^2} = 0 \Rightarrow \frac{\partial c_A(t, x)}{\partial x} = 0 \text{ for } x = 0 \text{ and } t \geq 0 \text{ (i.e., “all } t\text{”)}$$

Choose and name the dimensionless variables.

x_1 0 $-x_1$

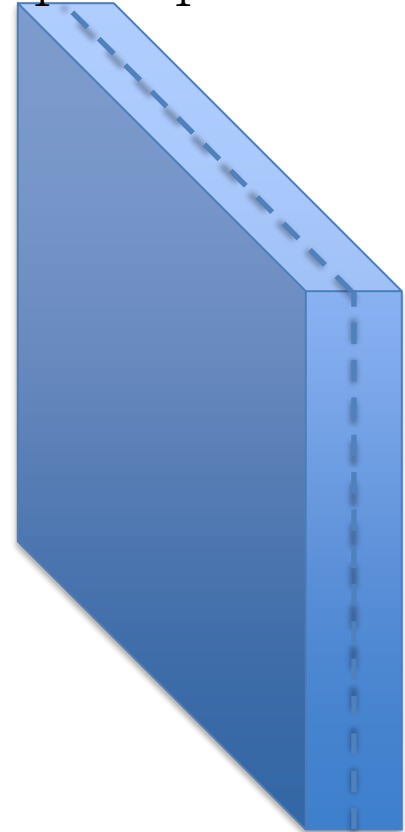
- Start with the dimensional equation and BCs:

$$c_A(0, x) = c_{A0} \quad \frac{\partial c_A}{\partial t} = D_{AB} \frac{\partial^2 c_A}{\partial x^2} \quad c_A(t, x_1) = c_{A1} \quad \frac{\partial c_A(t, 0)}{\partial x} = 0$$

- For $c_A(t, x)$, could use c_A/c_{A0} , but a “deviation variable” is often better, like $Y \equiv \frac{(c_{A1} - c_A)}{(c_{A1} - c_{A0})}$, where $c_A = (c_{A1} - c_{A0}) Y + c_{A1}$
- For x , use $X \equiv \frac{x}{x_1}$, where $x = x_1 \cdot X$
- To get normalizing form $\theta = t/?$, substitute & solve the rest:

$$\frac{(c_{A1} - c_{A0})}{1} \frac{\partial Y}{\partial t} = D_{AB} \frac{(c_{A1} - c_{A0})}{x_1^2} \frac{\partial^2 Y}{\partial x^2} \Rightarrow \frac{\partial Y}{\partial (D_{AB} t / x_1^2)} = \frac{\partial Y}{\partial \theta} = \frac{\partial^2 Y}{\partial x^2}$$

- And $Y(0, X) = 1$; $Y(\theta, 1) = 0$; $\frac{\partial Y(\theta, 0)}{\partial x} = 0$ where $\theta = \frac{D_{AB} t}{x_1^2}$



Example: Pure reactant is fed into an isothermal tubular reactor with steady-state diffusion, convection, and 1st-order isomerization $A \rightarrow B$ (molar consumption rate of A is $-kC_A$ mol/s /m³). Assume C_A is a function only of axial distance z . There is no change in the molar flow rate due to reaction, so molar density is constant.

[In "plug flow", there would be no axial diffusion.]

Write the species continuity equation for A and a boundary condition for $C_A(z=0)$, and then de-dimensionalize them.

Cylindrical coordinates:

$$\begin{aligned}
 & \cancel{\frac{\partial C_A}{\partial t}} + \left(\cancel{v_r \frac{\partial C_A}{\partial r}} + \cancel{v_\theta \frac{1}{r} \frac{\partial C_A}{\partial \theta}} + v_z \frac{\partial C_A}{\partial z} \right) \\
 & \quad \text{Steady state} \qquad \qquad \qquad C_A(z \text{ only}) \\
 & \quad \quad \quad = \mathcal{D}_{AB} \left(\cancel{\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial C_A}{\partial r} \right)} + \cancel{\frac{1}{r^2} \frac{\partial^2 C_A}{\partial \theta^2}} + \frac{\partial^2 C_A}{\partial z^2} \right) + \cancel{R_A} \quad (-kC_A)
 \end{aligned}$$